

# Development of a generic method to describe care pathways of patients with rare diseases using the French National Health Data System (SNDS).

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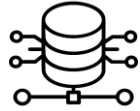
(3) Medical Assessment and Information Department, Université de Lorraine, CHRU-Nancy, Nancy, 54000, France

(4) L'institut des Maladies Génétiques Imagine, Paris, France

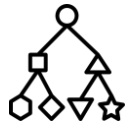
# Outline

- 1 **Introduction:** value of embeddings for claims data
- 2 **Method :** Computing, using and evaluating embeddings to identify *medical care events*
- 3 **Results :** Comparison of embeddings quality on different populations
- 4 **Application :** identifying *medical care events* for a rare disease
- 5 **Discussion and conclusion**

# French National Health Data System (SNDS)



Pseudonymized medical claim forms database.



All care events are coded according to classification systems (ICD-10 , ATC , biological acts, medical devices and medical procedure).



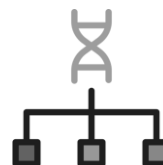
## Some numbers about the SNDS:

- **1,2 billion** medical claim forms
- **450 TB** of data
- Care pathway of **99%** of the population since 2010

1

# BNDMR (French National Rare Diseases Registry)

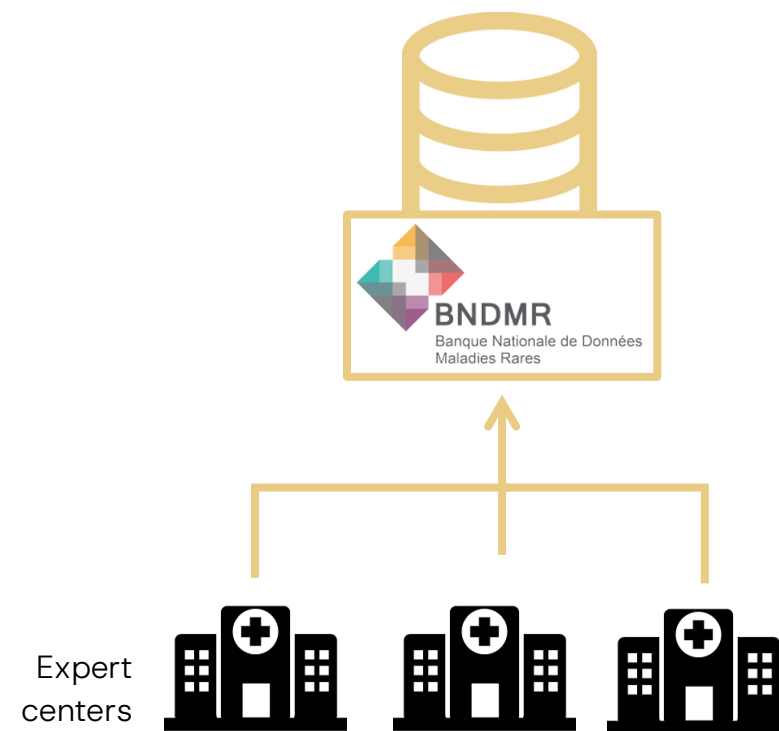
RD diagnosis: **orpha code** (Down syndrome = 870,  
cystic fibrosis = 586, ...)



Data collection in **expert centers** (2300+)



Enable to identify **cohort** of patients with RD



1

# Drómos study: linkage between SNDS and BNDMR

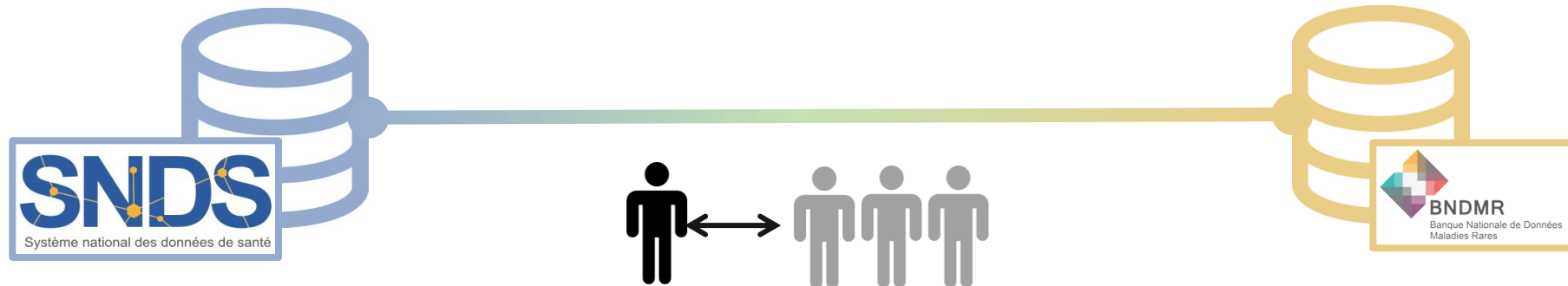


DROMOS

August 2023, ~ 350,000 patients linked

Historical data covering 2010–2022

3 controls per case ; matched on (gender, age, city of residence)

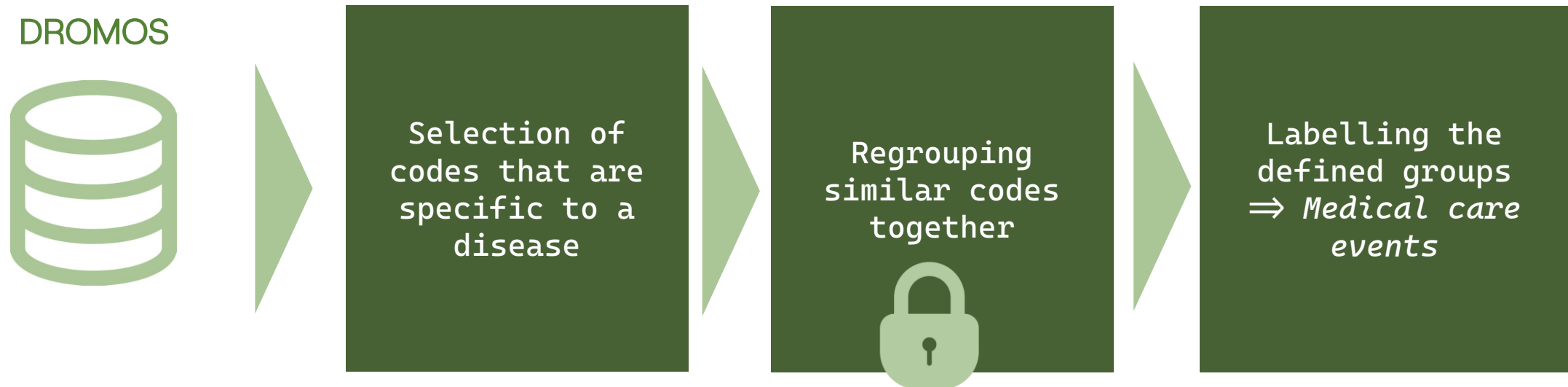


Objective of the linkage: Identify the care pathway of each rare disease



# 1 Pipeline in one picture :

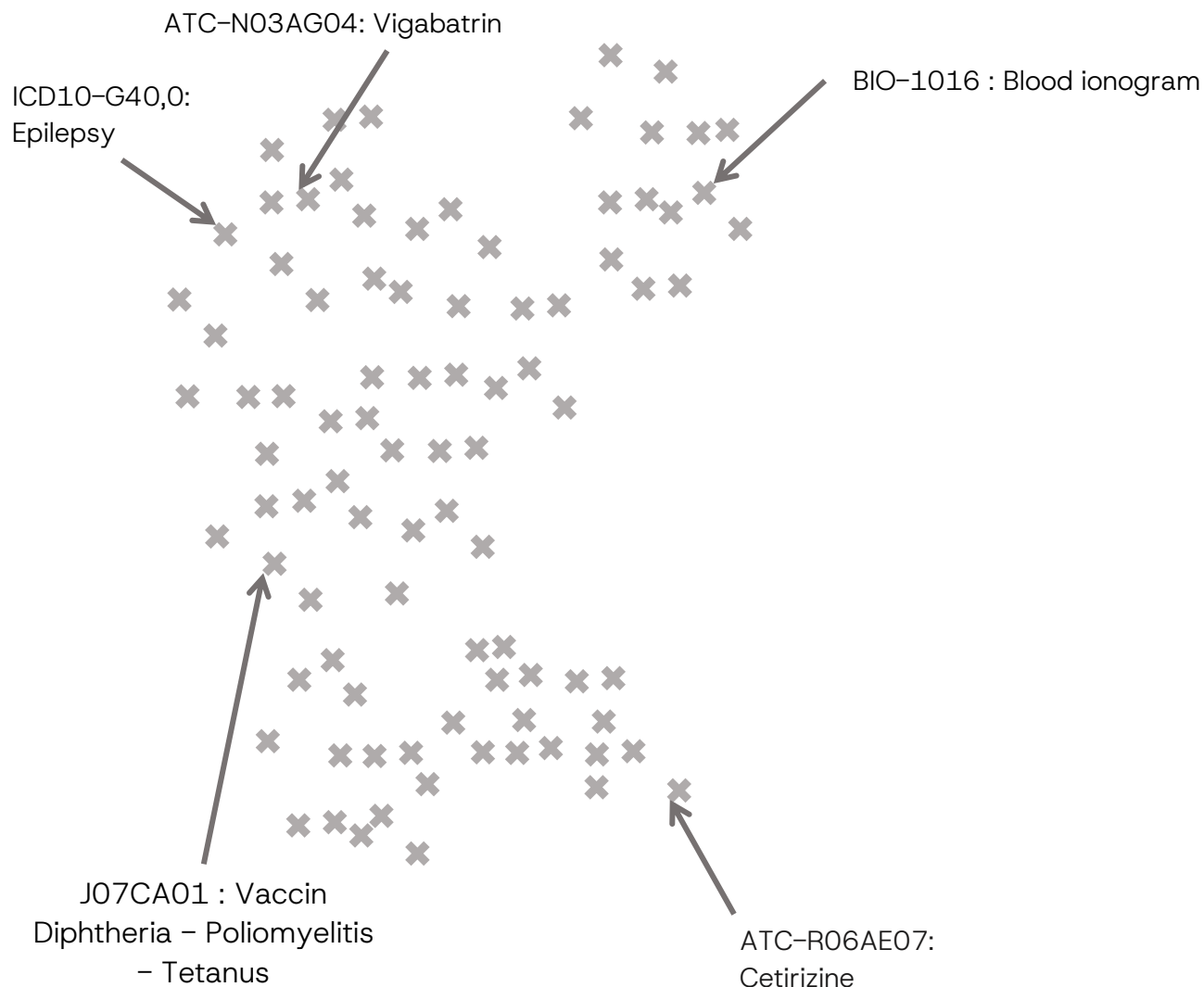
Methodological challenge : Build generic unsupervised method, to simplify the real-world claim data into a few **medical care events** that characterize a disease.



*How to group medical codes?*

1

# Embedding geometry relies on the input corpus



$\text{sim}_{\text{cosinus}} = [0;1]$

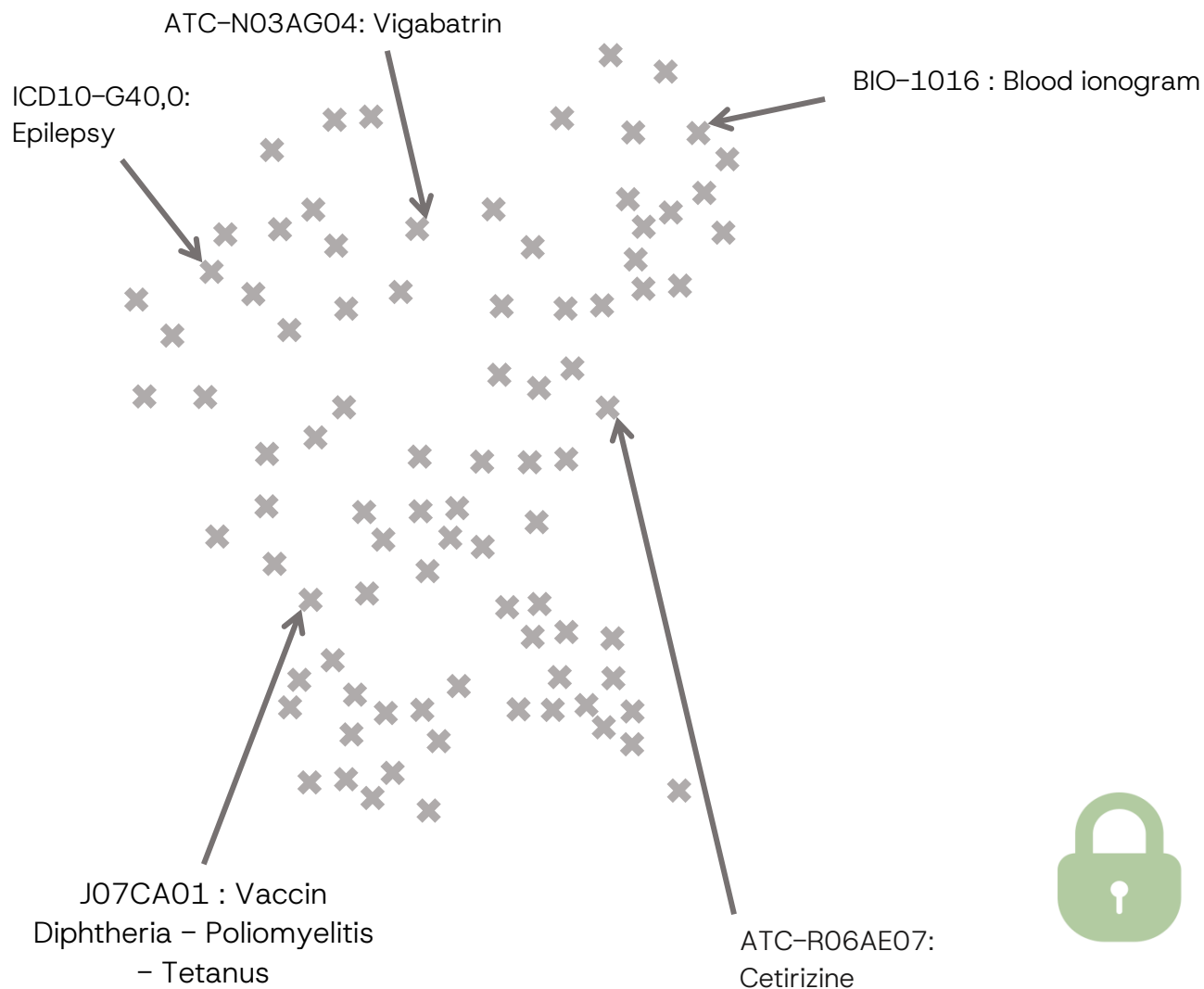
**Embedding A :**

$\text{sim}_{\text{cosinus}}(\text{Vigabatrin}, \text{Epilepsy}) = 0,77$

$\text{sim}_{\text{cosinus}}(\text{Vigabatrin}, \text{Cetirizine}) = 0,03$

## 1

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**Embedding B :**

$\text{sim}_{\text{cosinus}}(\text{Vigabatrin}, \text{Epilepsy}) = 0,55$

$\text{sim}_{\text{cosinus}}(\text{Vigabatrin}, \text{Cetirizine}) = 0,26$



The impact of the input corpus on the geometry of the embedding space.

## 2

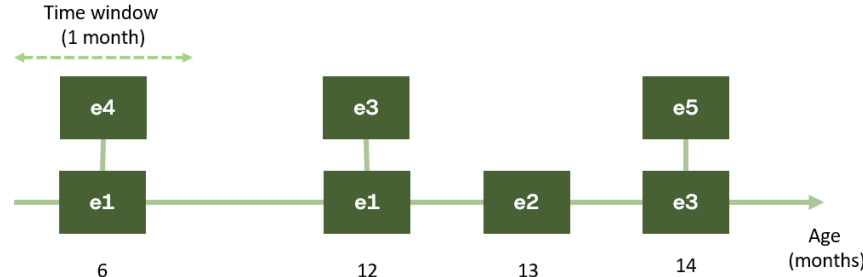
# Embedding construction method

General population



Specific population

- Pediatric RD
- 0 to 5 years old follow-up :



Example of an individual sequence

|    | e1 | e2 | e3 | e4 | e5 |
|----|----|----|----|----|----|
| e1 | 0  | 1  | 1  | 1  | 0  |
| e2 | 1  | 0  | 2  | 0  | 1  |
| e3 | 1  | 2  | 0  | 0  | 1  |
| e4 | 1  | 0  | 0  | 0  | 0  |
| e5 | 0  | 1  | 1  | 0  | 0  |

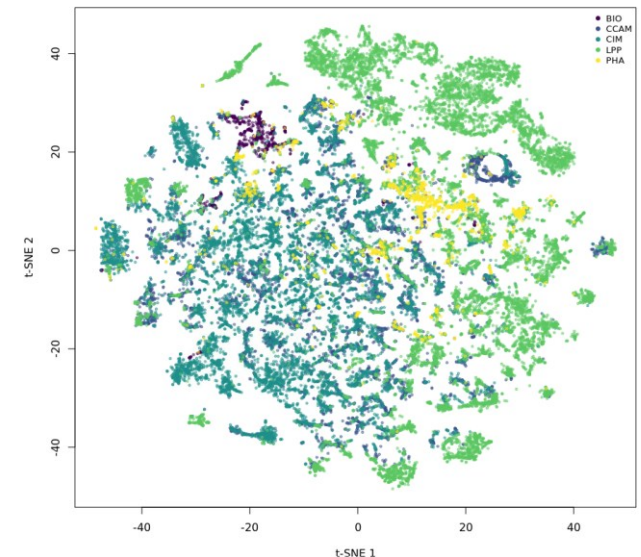
Corresponding co-occurrence matrix

- Co-occurrence matrix
  - Normalization using the positive mutual information matrix (PMI)
  - Dimension reduction via k-singular value decomposition
  - Calculation of embeddings
- Faujour, 2026

k dimensions

|    | V1          | V2           | ... | Vk-1        | Vk          |
|----|-------------|--------------|-----|-------------|-------------|
| e1 | 0.12        | 1.33         | ... | -0.92       | -2.54       |
| e2 | <b>0.52</b> | <b>-2.02</b> | ... | <b>0.75</b> | <b>1.21</b> |
| e3 | 2.36        | -0.78        | ..  | 1.49        | 0.1         |
| e4 | <b>0.56</b> | <b>-1.50</b> |     | <b>0.63</b> | <b>1.22</b> |
| e5 | -1.56       | 2.12         | ... | 0.01        | -0.50       |

X SNDS codes



2

## Embedding input corpus choice.

Best performing embedding for the population from which the medical care events are derived, or for a broader population?

### General population

- > 2 millions individuals
- > representative of the french population (Faujour, 2026)

or

### *Specific population*

- > *Pediatric RD*
- > *0-5 y.o. follow-up*



2

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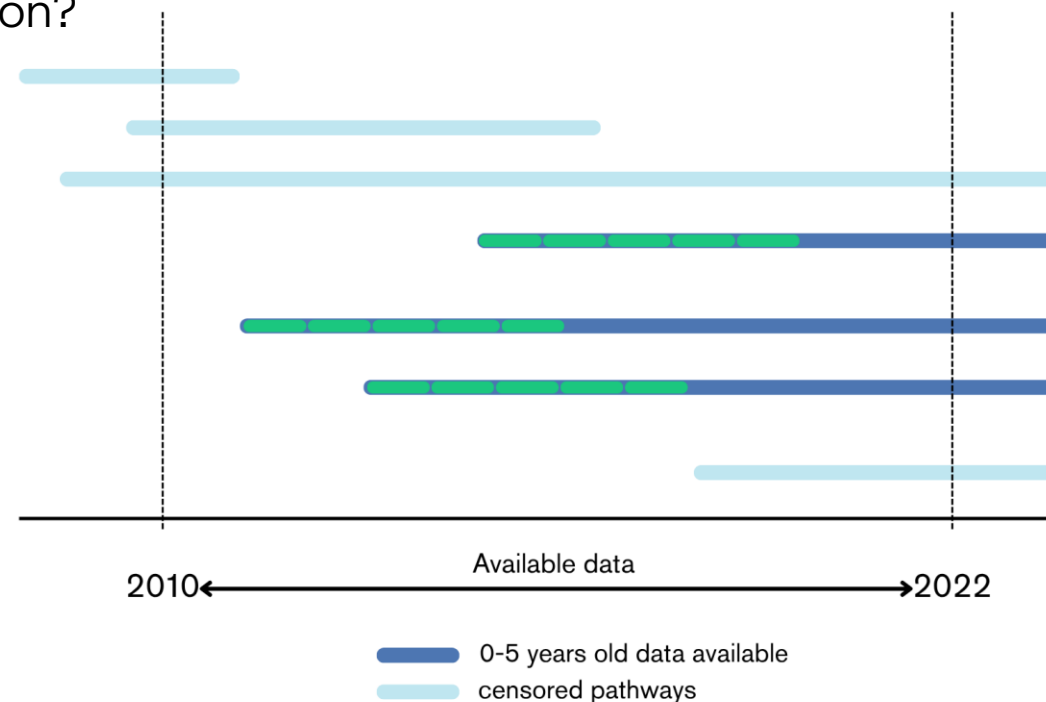
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| Set          | N token    | N case | N control | n SNDS codes<br>(> 10 occurrences) |
|--------------|------------|--------|-----------|------------------------------------|
| Set 1:[0-5]= | 27 552 890 | 25 823 | 76 903    | 9 586                              |



2

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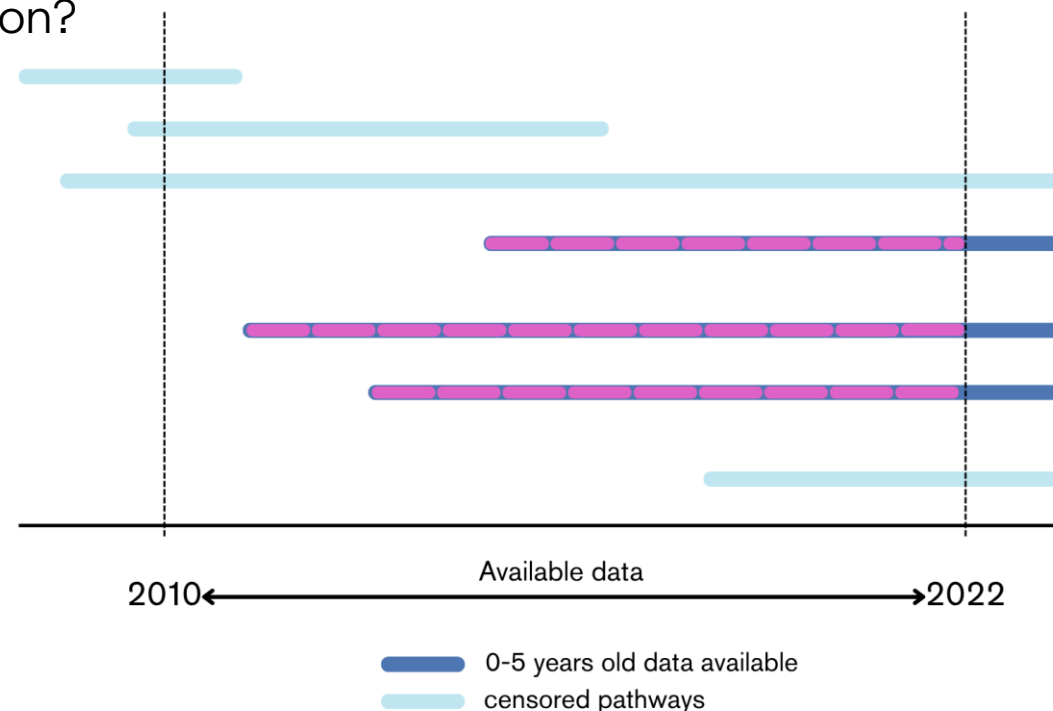
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| Set 1:[0-5]=  | 27 552 890 | 25 823 | 76 903    | 9 586                              |
| Set 2 :[0-5]+ | 42 003 366 | 25 823 | 76 903    | 12 366                             |



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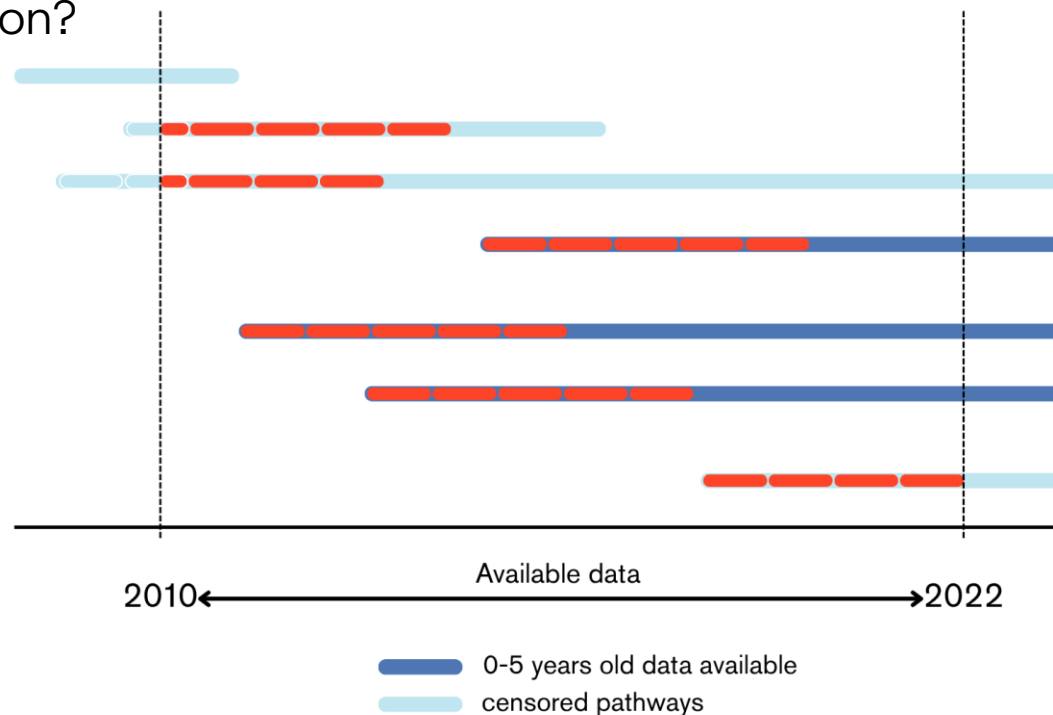
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| Set 3 :[0-5]- | 44 358 955 | 57 580 | 165 166   | 11 449                          |



2

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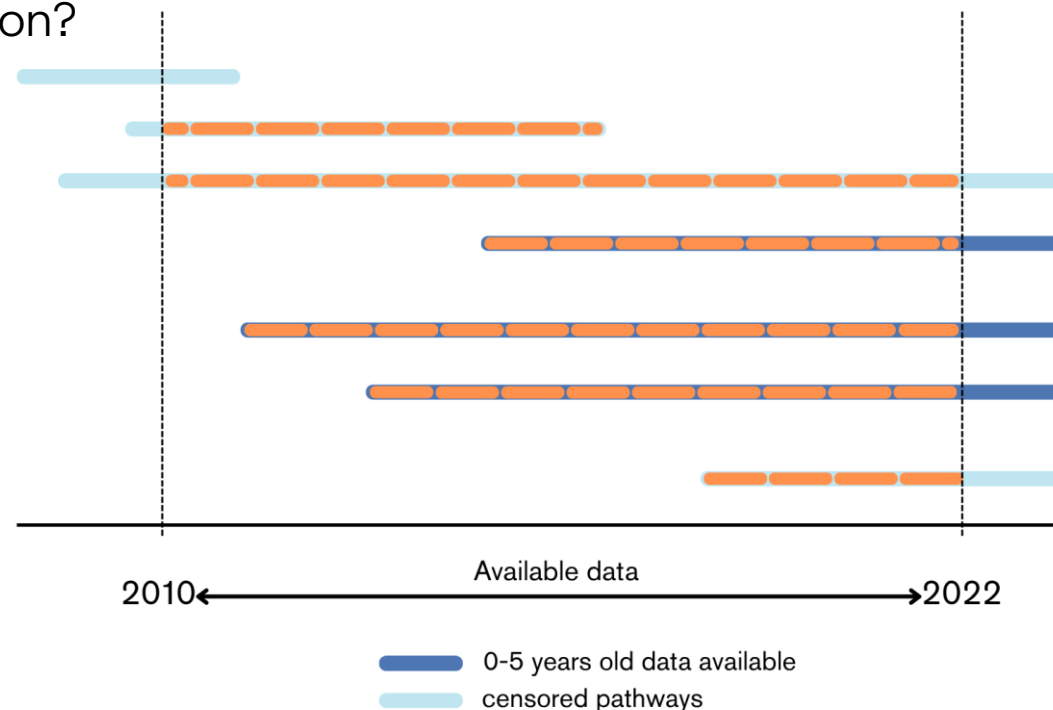
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| Set 4 : all   | 207 625 658 | 102 600 | 292 747   | 24 843                             |



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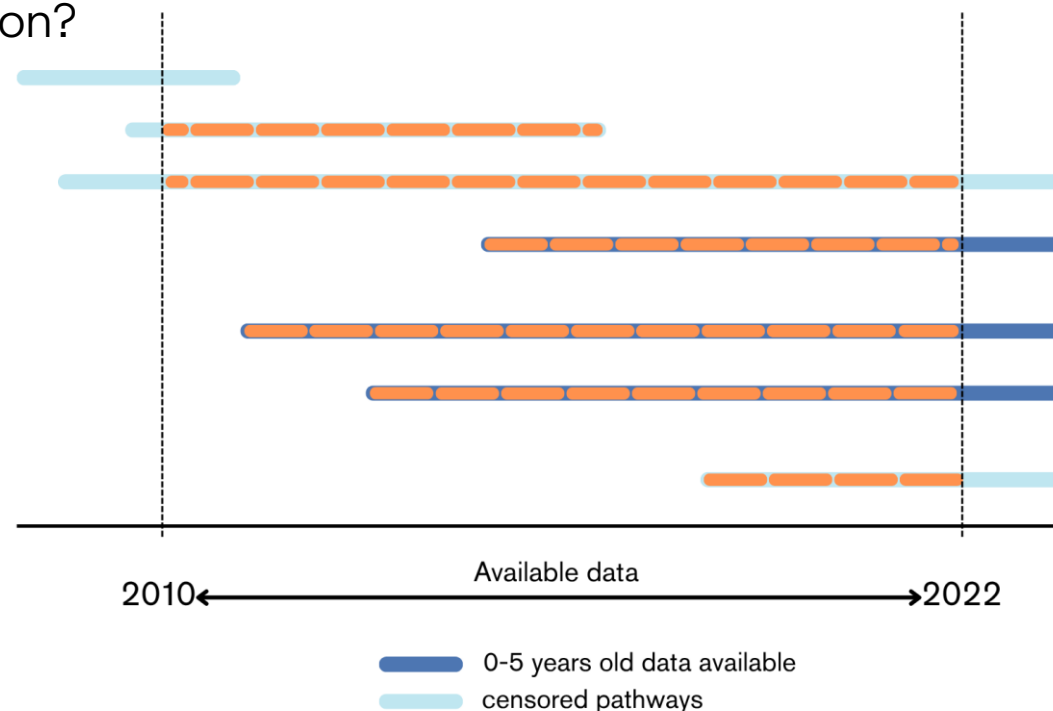
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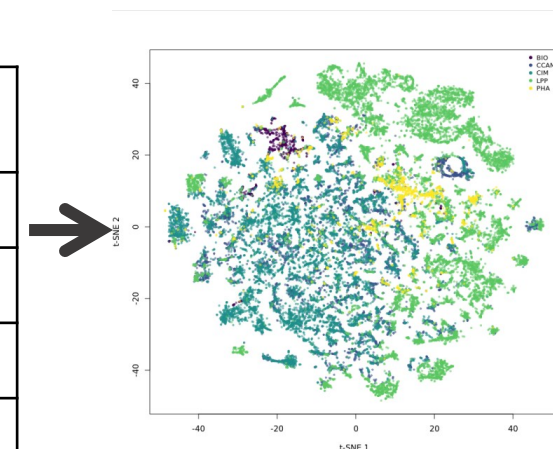
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## 2

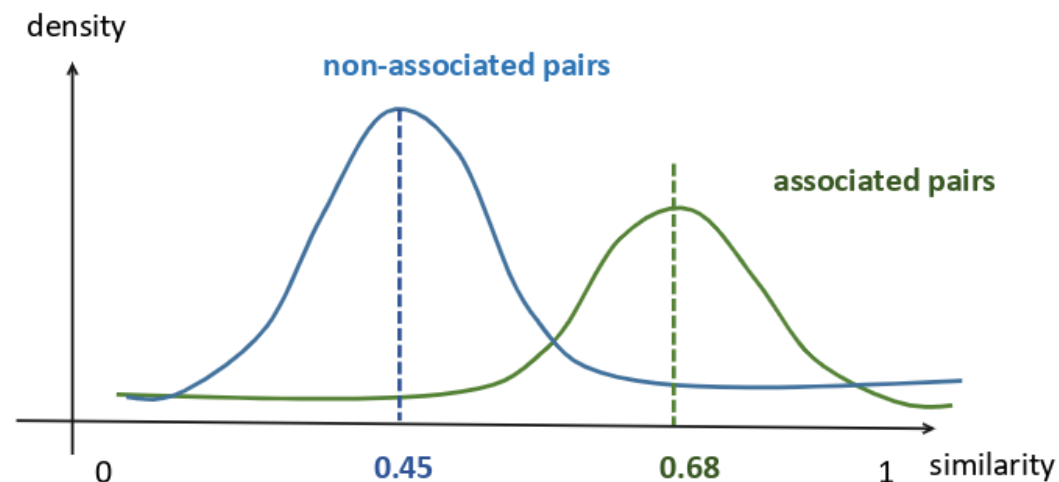
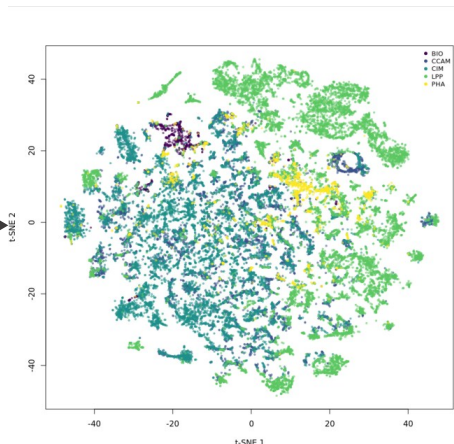
# Evaluation of the relevance of the embedding space

For pairs of sibling codes (ATC or ICD-10) :

**Associated pairs** : both codes show a statistically significant association with rare-disease patients ( $\chi^2$  test)

**Non-associated pairs** : codes that are not associated to the patients

|                |
|----------------|
| Set 1: [0-5]=  |
| Set 2 : [0-5]+ |
| Set 3 : [0-5]- |
| Set 4: all     |



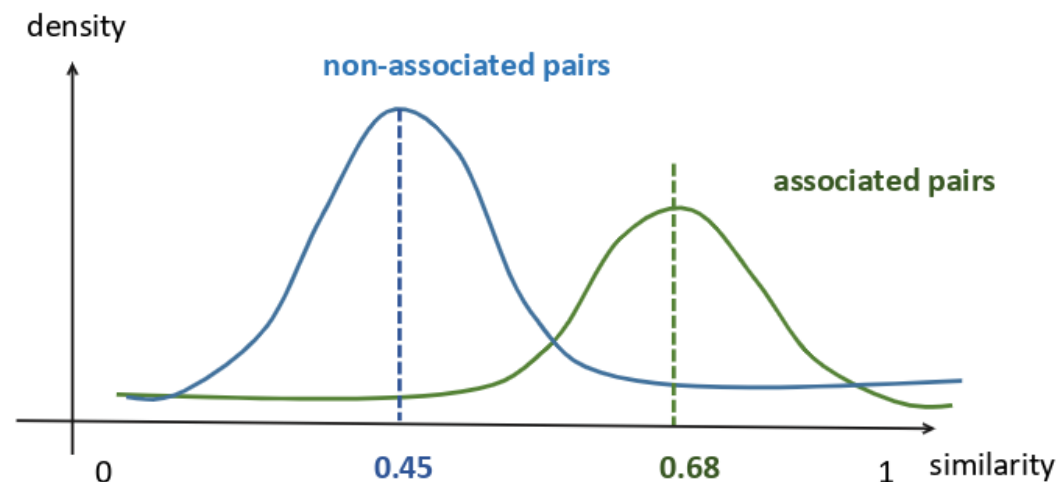
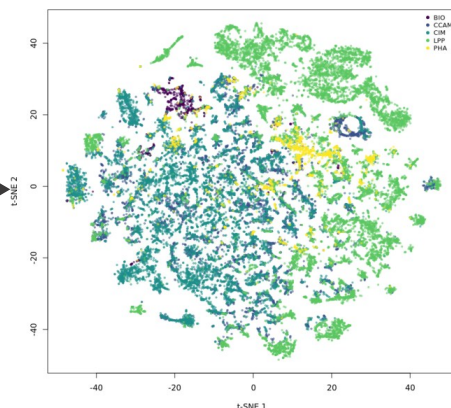
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|               |
|---------------|
| Set 1: [0-5]= |
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| Set 3: [0-5]- |
| Set 4: all    |



Evaluation 1: Clinical coherence: are RD-associated code pairs more similar than non-RD-associated code pairs?

$\rightarrow \Delta \text{sim} = \text{mean}(\text{sim}(\text{associated})) - \text{mean}(\text{sim}(\text{non-associated})) > 0$

Evaluation 2 : Disease specificity: is this difference larger in the disease-specific embedding than in the general-population embedding?

$\rightarrow \Delta \text{sim}_{\text{specific pop}} > \Delta \text{sim}_{\text{general pop}}$

$\rightarrow \text{Cohen's } d_{\text{specific pop}} > \text{Cohen's } d_{\text{general pop}}$



# Results : Comparison of embeddings quality on different populations

ICD-10

T-test

Evaluation 1 :  $\Delta\text{sim} > 0$ 

Evaluation 2 : strongest difference for set 4

|                  | Mean diff<br>Mean(sim <sub>associated</sub> ) - Mean (sim <sub>not associated</sub> ) | P value     | Cohen's d             |
|------------------|---------------------------------------------------------------------------------------|-------------|-----------------------|
| Emb set 1 [0-5]= | + 0.025 (0.373- 0.348)                                                                | 2.5e-7 ***  | 0.101 [0.061 ; 0.139] |
| Emb set 2 [0-5]+ | + 0.053 (0.415- 0.362)                                                                | 2.2e-16 *** | 0.206 [0.170 ; 0.242] |
| Emb set 3 [0-5]- | + 0.052 (0.411- 0.359)                                                                | 2.2e-16 *** | 0.210 [0.174; 0.246]  |
| Emb set 4 all    | + 0.125 (0.551- 0.425)                                                                | 2.2e-16 *** | 0.431 [0.452 ; 0.511] |
| Emb pop gen      | + 0.068 (0.490- 0.422)                                                                | 2.2e-16 *** | 0.250 [0.215 ; 0.286] |

ATC

|                  | Mean diff<br>Mean(sim <sub>associated</sub> ) - Mean (sim <sub>not associated</sub> ) | P value      | Cohen's d             |
|------------------|---------------------------------------------------------------------------------------|--------------|-----------------------|
| Emb set 1 [0-5]= | + 0.011 (0.480- 0.469)                                                                | 0.61         | 0.040 [-0.128; 0.221] |
| Emb set 2 [0-5]+ | + 0.075 (0.508- 0.433)                                                                | 0.001**      | 0.288 [0.119; 0.458]  |
| Emb set 3 [0-5]- | + 0.046 (0.518- 0.472)                                                                | 0.038**      | 0.179 [0.009; 0.349]  |
| Emb set 4 all    | + 0.154 (0.659 - 0.504)                                                               | 8.33e-11 *** | 0.562 [0.402; 0.721]  |
| Emb pop gen      | + 0.074 (0.596- 0.522)                                                                | 8.21e-4 ***  | 0.266 [0.105; 0.427]  |

## 4

## Application: Dravet syndrom main medical care events

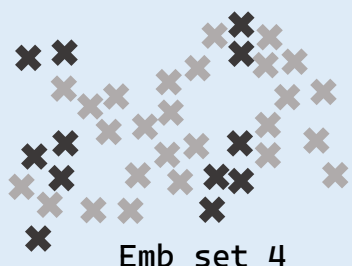


137

cases

399

controls



Emb set 4

1854

SNDS codes in total

235

Case-specific code

Hierarchical clustering

(ARI 0.75+/- 0.06)

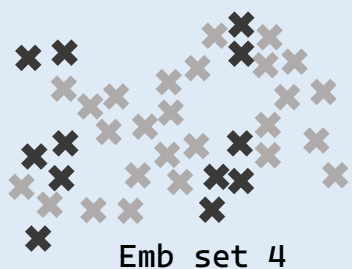
&gt; 15 clusters

# Application: Dravet syndrom main medical care events



137  
cases

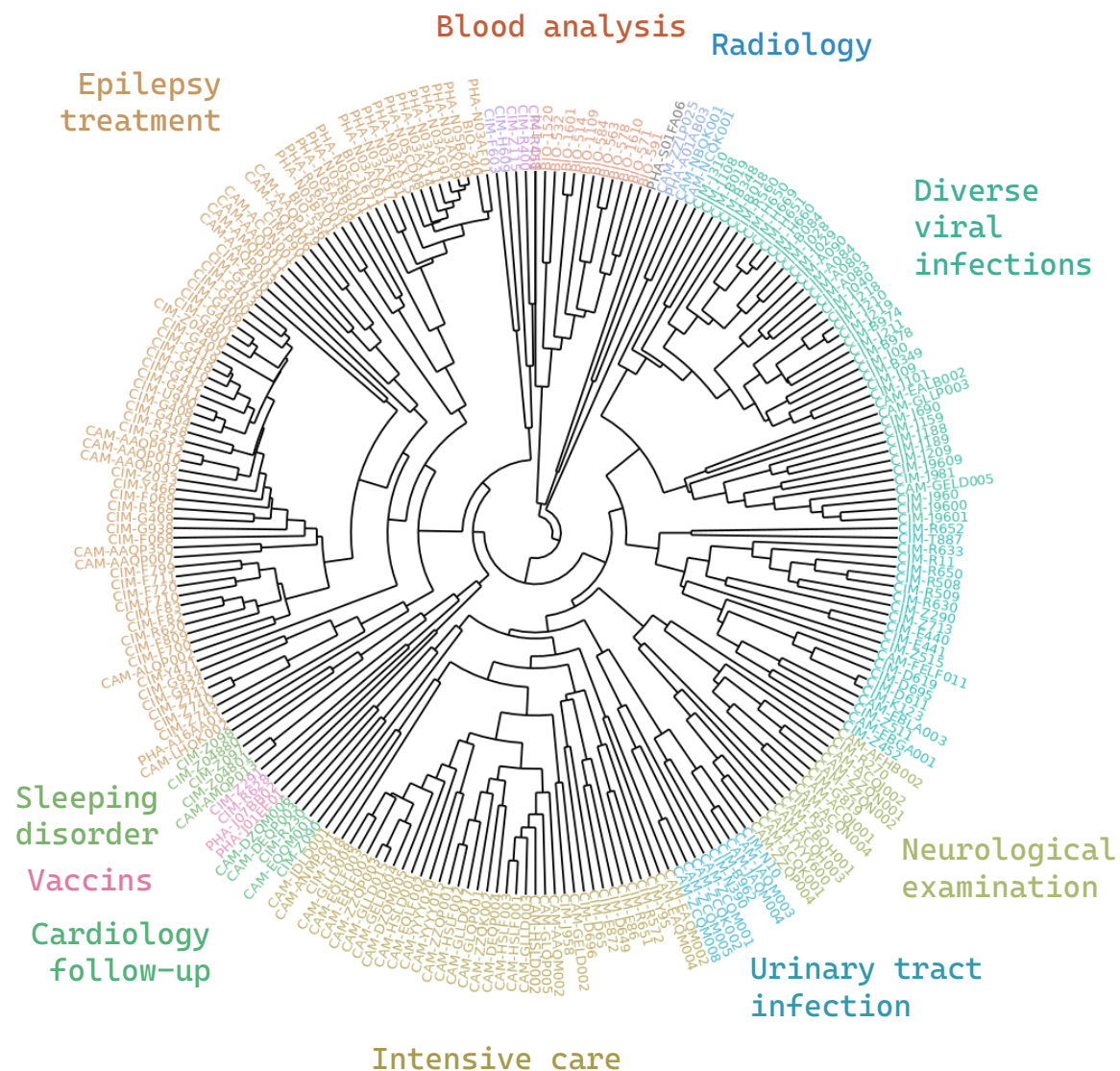
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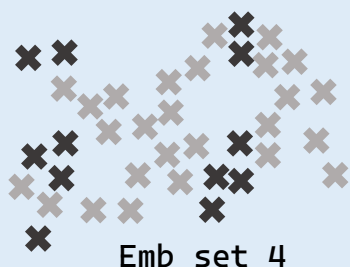
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# Application: Dravet syndrom main medical care events



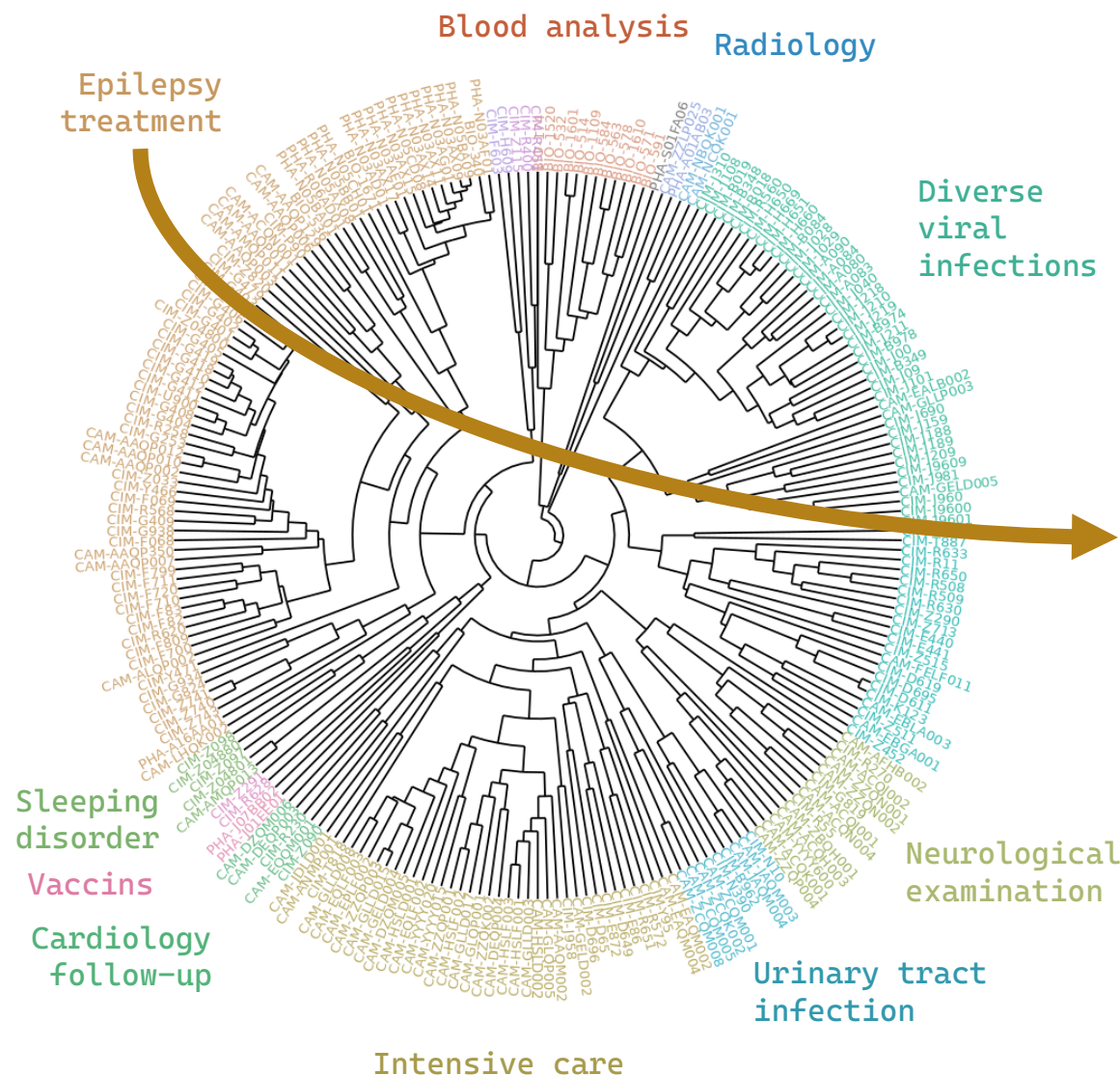
137 cases  
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> 15 clusters



CAM-AAQP002, CAM-AAQP003, CAM-AAQP006, CAM-AAQP007, CAM-AAQP010, CAM-AAQP011, CAM-AAQP012, CAM-AAQP129, CAM-AAQP350 : Electroencephalography / EEG procedures

• CAM-ACQN001, CAM-ALQP002, CAM-AMQP011, CAM-LHQK002 : Other neurological diagnostic procedures

• BIO-1671 : Antiepileptic drug not specifically listed in NABM (blood)

• BIO-340 : Valproic acid (Depakine) (blood)

• A16A : Other alimentary tract and metabolism products

A16AA01 (Levocarnitine)

• N03A : Antiepileptic drugs

N03AD01 (Ethosuximide), N03AE01 (Clonazepam), N03AF01 (Carbamazepine), N03AG01 (Valproic acid), N03AX09 (Lamotrigine), N03AX11 (Topiramate), N03AX14 (Levetiracetam), N03AX15 (Zonisamide), N03AX17 (Stiripentol)

• N05A : Antipsychotics

• N05B : Anxiolytics

• N05C : Hypnotics and sedatives

• N05CH : Melatonin receptor agonists

• R06A : Antihistamines for systemic use

• G40 : Epilepsy and recurrent seizures : G400, G401, G402, G403, G404, G405, G406, G408, G409

• G41 : Status epilepticus : G410, G411, G412, G418, G419

• G82 : Paralytic syndromes : G824 (Spastic quadriplegia)

• G93 : Other disorders of brain : G934, G938

...

• P90 : Convulsions of newborn : P90

• R25 : Abnormal involuntary movements : R258

• Z03 : Medical observation for suspected diseases : Z033

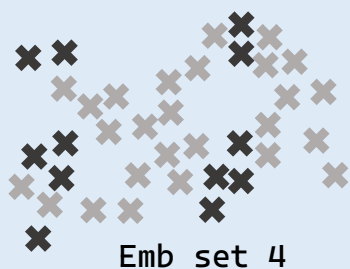
• Z04 : Examination and observation for other reasons : Z04800

# Application: Dravet syndrom main medical care events



137  
cases

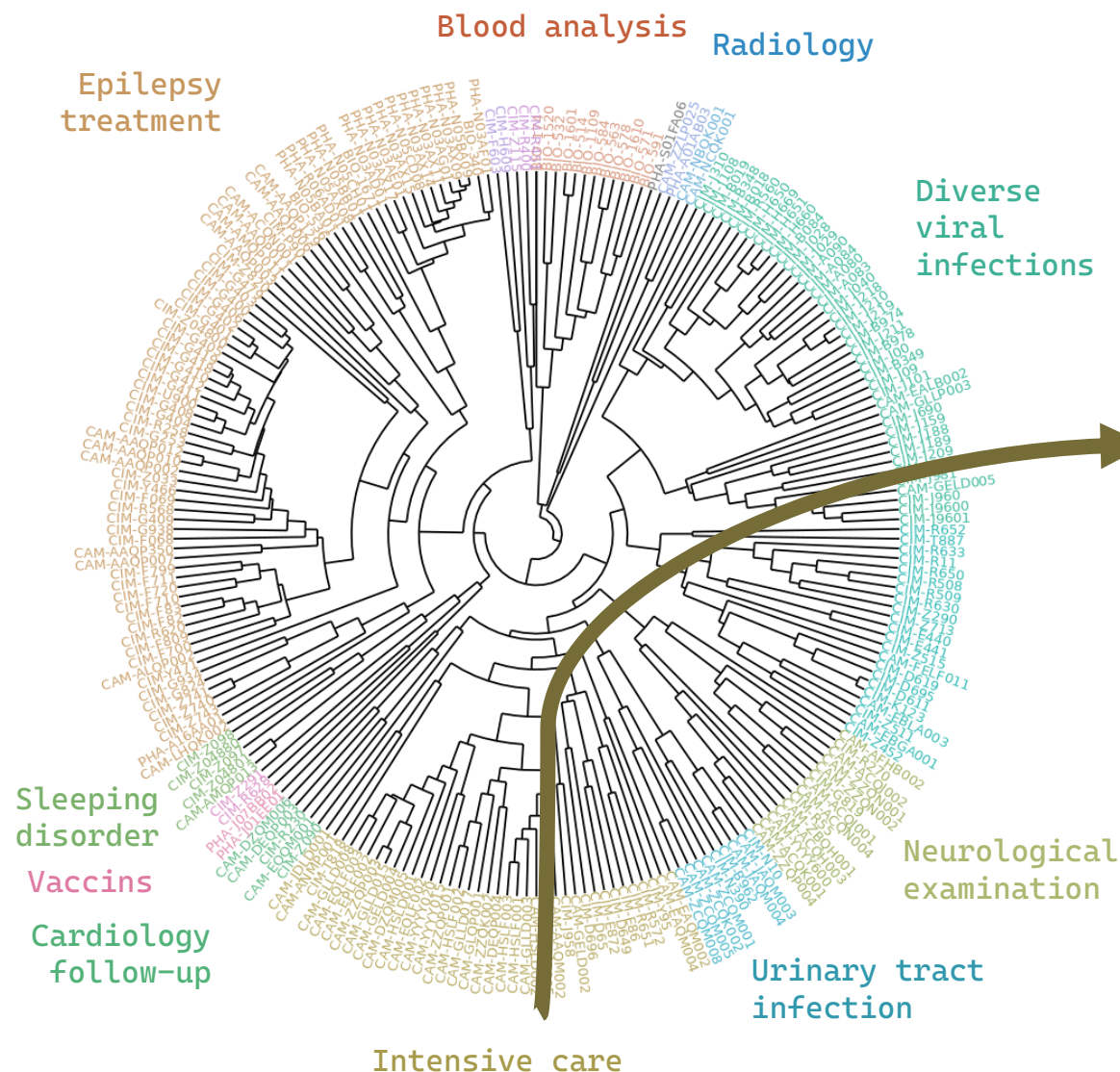
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- CAM-AAQM002 : Transfontanellar brain ultrasound
- CAM-DEQP004, CAM-DZQM005, CAM-EAQM002, CAM-EAQM004 : Cardiac and neurological monitoring / ultrasound procedures
- CAM-EPLF002, CAM-EQLF002, CAM-EQLF003 : Vascular access and intravenous therapies
- CAM-GELD002, CAM-GELD004 : Tracheal intubation
- CAM-GLLD003, CAM-GLLD008, CAM-GLLD015, CAM-GLLD017, CAM-GLQP001, CAM-GLQP005 : Respiratory support and oxygen monitoring
- CAM-HSLD001, CAM-HSLD002, CAM-HSLF001, CAM-HSLF002, CAM-HSLF003 : Enteral and parenteral nutrition
- CAM-JDLD001 : Indwelling urinary catheterization
- CAM-YYYY015, CAM-YYYY020 : Intensive care packages
- CAM-ZZQK002 : Bedside X-ray
- CAM-ZZQP001, CAM-ZZQP003 : Medical monitoring during intrahospital transport

- CIM-D649, CIM-D65, CIM-D696 : Hematological disorders
- CIM-E86, CIM-E872 : Hypovolemia and acidosis
- CIM-J958, CIM-T818 : Post-procedural complications
- CIM-R572, CIM-R651, CIM-Y95 : Infectious, inflammatory and nosocomial complications

Intensive care

## 4 Discussion

### Limitations :



**SNDS** : Partial clinical picture,  
only healthcare claim data

### Next steps :

Apply the methodology to a **large number of rare pediatric diseases**  
> Panorama for pediatric rare diseases



**Feature engineering** remains  
an important methodological  
step.



## 4 Conclusion



- A generic, unsupervised approach : SNDS codes > “medical care events”



- Real-world data for rare pediatric diseases : help patients and clinicians



- Leverages the properties of the SNDS data: large population, sparse events



# Thank you for your attention



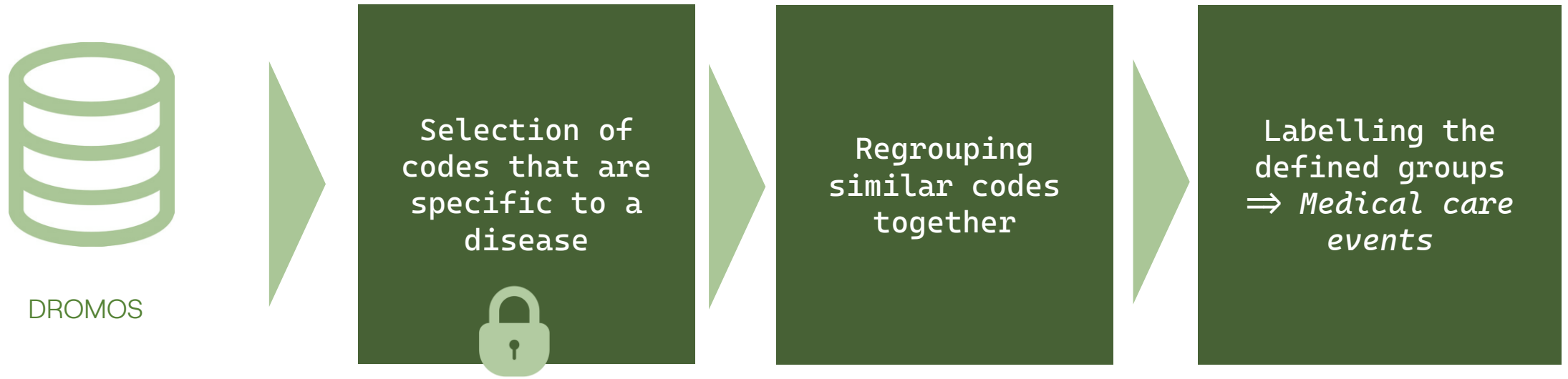
M4DI project



HeKA team

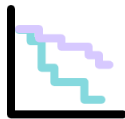


Renée Le Clech  
[renee.le-clech@inria.fr](mailto:renee.le-clech@inria.fr)



### Case specific codes

- Filter code >5% of the controls
- Statistical tests on the SNDS codes :
  - o Chi 2
  - o Wilcoxon
  - o Kaplan Meier



SNDS Codes that are specific to the cases in proportion, quantity and in timing

Methodological stability  
*Does this approach produce consistent medical care events?*



Bootstrapping: resampling  
the cases and controls



Bootstrap #1



Bootstrap #2



Bootstrap #3

K-means vs AHC



Adjusted Rand  
Index (ARI)

